MILITARY SPECIFICATION

INDICATORS, PRESSURE, SYNCHRO, DUAL, GENERAL PURPOSE

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design requirements and all performance requirements for the procurement of hermetically sealed, clamp-mounted, synchro, dual pressure indicators for operation on 26 volts, 400 cycles per second.

1.2 Classification - The indicators shall be of the following MS designations with the case, dial scale, pointers marked, and with or without integral lighting as shown for the applicable MS part number specified (see 6.2).

Without Integral Lighting				With Integral Lighting			
MS Part No.	Case Size	Dial Scale	Pointers Marked	MS Part No.	Case Size	Dial Scale	Pointers Marked
MS25477- 1	2-inch	0- 5	1, 2	MS25477- 2	2-inch	0- 5	1, 2
- 3		0- 5	3, 4	- 4		0- 5	3, 4
- 5		0-10	1, 2	- 6		0-10	1, 2
- 7		0-10	3, 4	- 8		0-10	3, 4
- 9		0-20	1, 2	-10		0-20	1, 2
-11		0-20	3, 4	-12		0-20	3, 4
-13		0-25	1, 2	-14		0-25	1, 2
-15		0-25	3, 4	-16		0-25	3, 4
-17		0-30	1, 2	-18		0-30	1, 2
-19		0-30	3, 4	-20		0-30	3, 4
MS25477-21	2-inch	0-35	1, 2	MS25477-22	2-inch	0-35	1, 2

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Without Integral Lighting				With Integral Lighting			
MS Part No.	Case Size	Dial Scale	Pointers Marked	MS Part No.	Case Size	Dial Scale	Pointers Marked
MS25477-23	2-inch	0-35	3, 4	MS25477-24	2-inch	0-35	3, 4
-25		0-40	1, 2	-26		0-40	1, 2
-27	2-inch	0-40	3, 4	-28	2-inch	0-40	3, 4
-29	3-inch	0- 5	1, 2	-30	3-inch	0- 5	1, 2
-31		0- 5	3, 4	-32		0- 5	3, 4
-33		0-10	1, 2	-34		0-10	1, 2
-35		0-10	3, 4	-36		0-10	3, 4
-37		0-20	1, 2	-38		0-20	1, 2
-39		0-20	3, 4	-40		0-20	3, 4
-41		0-25	1, 2	-42		0-25	1, 2
-43		0-25	3, 4	-44		0-25	3, 4
-45		0-30	1, 2	-46		0-30	1, 2
-47		0-30	3, 4	-48		0-30	3, 4
-49		0-35	1, 2	-50		0-35	1, 2
-51		0-35	3, 4	-52		0-35	3, 4
-53		0-40	1, 2	-54		0-40	1, 2
MS25477-55	3-inch	0-40	3, 4	MS25477-56	3-inch	0-40	3, 4

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen
DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors and Other Uses
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)

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SPECIFICATIONS

Federal (Continued)

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating, Electrodeposited, Requirements for

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-T-5350	Transmitters, Synchro, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-T-7748	Transmitters, Pressure, Synchro, Aircraft, 320-Degree Movement, High Temperature General Purpose
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

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SPECIFICATIONS

Military (Continued)

MIL-S-20708	Synchros, 60 and 400 Cycle, General Specification
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionics Equipment, General Specification for
MIL-L-25467	Lighting, Integral, Instrument, General Specification for
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED. STD. No. 595	Colors

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS3116	Connectors, Plug, Electric, Solder Type, Straight, Bayonet Coupling
MS24367	Lamp, Incandescent - Miniature Integral Lighting
MS24515	Lamp - Sub Miniature

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STANDARDS

Military (Continued)

MS25477	Indicators, Pressure, Synchro, Dual, General Purpose
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33586	Metals, Definition of Dissimilar

2.2 Availability of documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

* 3.1 Preproduction - The indicators furnished under this specification shall be products, samples of which have been tested and have passed the preproduction tests specified herein.

3.2 Selection of government documents - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts - Commercial parts having suitable properties shall be used, where on the date of invitations for bids, there are no suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts, cotter pins, having suitable properties, may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

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3.2.2 Standard parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.3 Metals - Metals shall be of the corrosion resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray or atmospheric conditions to which the indicator may be subjected when in storage or during normal service life.

3.3.3.1 Dissimilar metals - Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings.

* 3.3.3.2 Magnesium alloy parts - Magnesium alloy parts shall be treated in accordance with Specification MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

3.3.3.3 Aluminum alloy parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Small holes, pipe threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

* 3.3.3.4 Iron and steel parts - Iron and steel parts which are in hermetically sealed cases shall be cadmium plated in accordance with Specification QQ-P-416. Parts which are not in sealed cases shall be chromium, nickel, or zinc plated in accordance with Specifications QQ-C-320, QQ-N-290, and QQ-Z-325, respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium plated.

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3.3.4 Protective treatment - When materials are used in the construction of the indicators that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip or scale during normal service life or are affected by extremes of atmospheric or environmental conditions, shall not be used.

3.3.5 Fungus-proof materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.6 Fumes and vapors - Materials used in the construction of the indicators shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.

* 3.4 Design and construction - The indicator shall be designed so that either one or two indicators can be operated by two transmitters conforming to Specification MIL-T-7748 or other suitable 320-degree movement synchro pressure transmitters conforming to Specification MIL-T-5350. Each indicator shall incorporate two synchro receivers geared or otherwise connected to two concentric pointers. The indicator shall be so constructed that no parts will work loose in service and will withstand the normal shocks, vibrations and such other conditions as are incident to service without failure, shipping, storage, and installation.

3.4.1 Indicator synchro - The indicator synchro shall consist of a salient two-pole wound rotor (primary) designed for operation on 26 volts, 400 cps, and a three-winding, Y-connected stator (secondary). The stator shall be wound so that the amplitudes of the induced voltages in the three windings vary sinusoidally with the angular position of the rotor.

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3.4.1.1 Primary currents - The current drawn by one indicator synchro shall not exceed 120 milliamperes, with secondaries (stator) open circuited.

3.4.1.2 Power input - The power consumption of each indicator synchro shall not exceed 0.60 watt.

3.4.1.3 Secondary voltage - The maximum open circuit voltage measured between any two secondary leads shall be 11.8 ±0.3 volt. The variation between the three maximum secondary voltages on any one indicator synchro shall not exceed 0.2 volt.

3.4.2 Maintenance - The design shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly using those tools and items of maintenance equipment which are normally available as commercial standards.

* 3.4.3 Case - The case shall conform to MS25477. The case size shall be as required for the applicable MS part number (see 1.2). The dimension designated as "L" on MS25477 shall not exceed 5 inches for each indicator. The case shall be made of non-ferrous, low density metal, uniform in texture, having a smooth surface, and shall be hermetically sealed. The case shall be finished with a lusterless black material Color No. 37038 of Fed. Std. No. 595. The finishing material shall be of a durable type to withstand usage encountered in service.

3.4.3.1 Hermetic sealing - The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed and replaced, and the case resealed at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any atmosphere to which the indicator may be subjected in normal use in military aircraft.

* 3.4.3.2 Filling medium - The filling medium shall be a mixture of 90 percent nitrogen and ten percent helium. The nitrogen used shall be in accordance with Specification BB-N-411, Type I, Class I, Grade C. The filling medium shall contain not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 ± 0.1 atmosphere.

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3.4.4 Weight - The weight of each 2-inch indicator shall not exceed 1.0 pound, and the weight of each 3-inch indicator shall not exceed 1.5 pounds

3.4.5 Screw threads - Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.5 Electrical requirements -

* 3.5.1 Primary input power requirements - The indicator shall be designed to operate on AC power, single phase, 26 ±2V, 400 ± 20 cps.

3.5.2 Undervoltage protection - The indicator shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage is within the specified limits.

* 3.5.3 Integral lighting - When specified, the indicator shall be provided with integral lighting in accordance with Specification MIL-L-25467.

* 3.5.4 Lamps - The lamps shall be located within the indicator case. Lamps conforming to MS24367 or MS24515 may be used in lieu of, or in addition to, the lamps specified in Specification MIL-L-25467.

3.5.5 Electric connector - The electric connector shall be mounted on the rear surface of the case with the axis of the connector within 1/16 inch of the axis of the case. The largest polarizing keyway of the electric connector shall be at the top center of the shell of the connector. The electric connector for each nonintegrally lit indicator shall conform to Specification MIL-C-26482 and shall mate with MS3116-16-8S. The electric connector for each integrally lit indicator shall conform to Specification MIL-C-26482 and shall mate with MS3116-12-10S.

* 3.5.6 Wiring - The electrical circuit of each indicator shall conform to Figure 1. Signal rotations A, D, C and A, G, F at the pins of the electric connector of the indicator shall produce increasing pressure indications. Internal wiring shall be color coded, neat, and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering and shall be insulated from the indicator case.

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3.5.6.1 Indicators without integral lighting - Each unlighted indicator shall be color coded and connected as shown in Figure 1.

* 3.5.6.2 Indicators with integral lighting - Each integrally lighted indicator shall be lighted in accordance with the requirements of Specification MIL-L-25467. The indicator shall be color coded and connected as shown in Figure 1 with the exception that two additional pins, "H" and "J", shall be required for the connector. The integral lighting lamps shall be connected between pins "H" and "J". There shall be no connection within the indicator between the integral lighting lamps and any other part of the electrical circuit of the indicator.

* 3.6 Dial - The dial shall be securely mounted in such a manner that it will not loosen or slip when the indicator is vibrated. If screws are used they shall be so located that they will not interfere with the proper location of any dial marking.

* 3.6.1 Visibility of dial - The pointer, numerals, at least 1/16th inch of the shortest graduations, and all other specified markings on the dial, with the exception of the electrical zero reference mark shall be visible from any point within the frustum of a cone whose side makes an angle of 30° with a perpendicular to the dial and whose small diameter is the aperture of the case.

* 3.6.2 Dial marking - The dial shall be marked as shown in Figure 2 and the applicable scale of Figure 3 as listed in 1.2 for the applicable MS part number. Unless otherwise specified, all visible portions of the dial shall be lusterless black Color No. 37038 of Fed. Std. No. 595 and all markings on the dial shall be lusterless white Color No. 37875 of Fed. Std. No. 595. All markings shall be durable to withstand usage encountered in service. There shall be no change in the color of the markings as a result of subjecting the indicator to the tests specified herein. The form of the numerals and letters shall be in accordance with MS33558. Numerals shall distinctly indicate the graduation to which each applies. If practicable, each numeral shall be placed so that the center of area of the numeral is on the radial line

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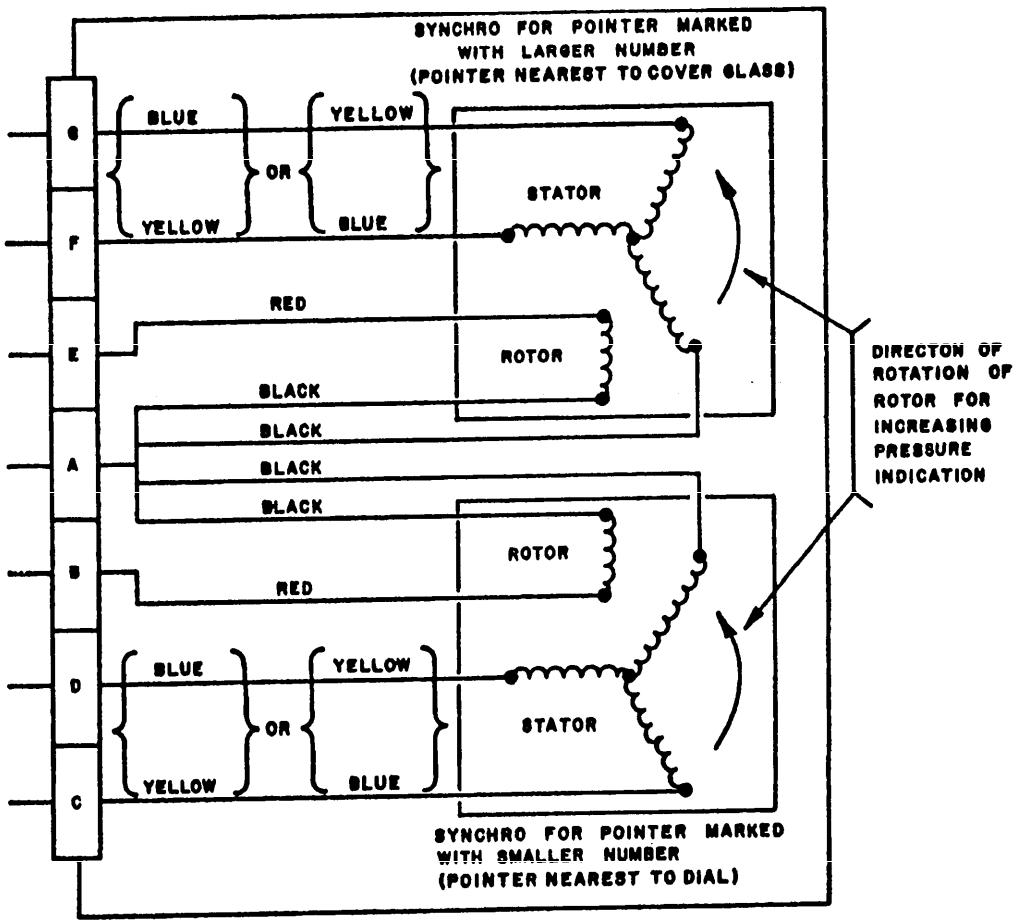


FIGURE 1
ELECTRICAL CIRCUIT FOR NON-LIGHTED INDICATOR

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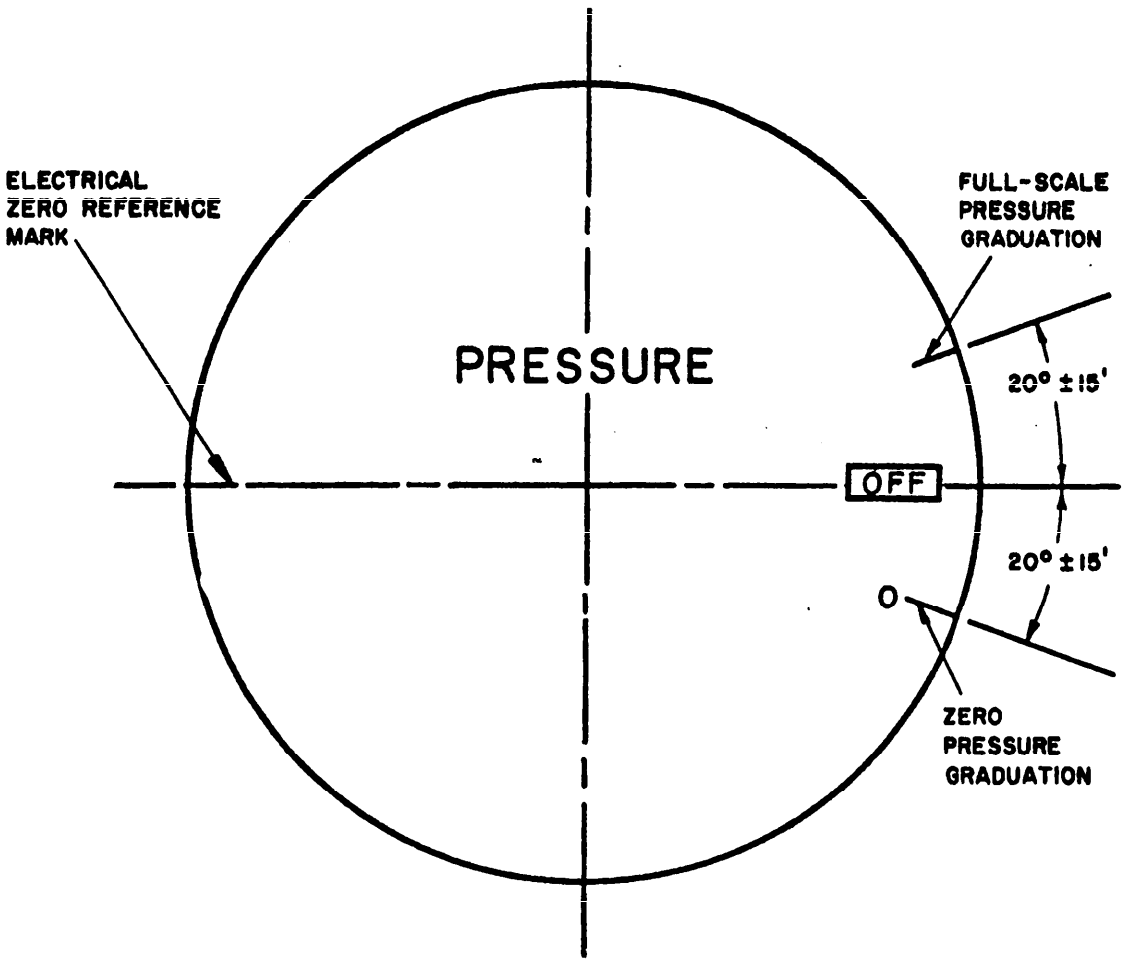


FIGURE 2 DIAL LAYOUT

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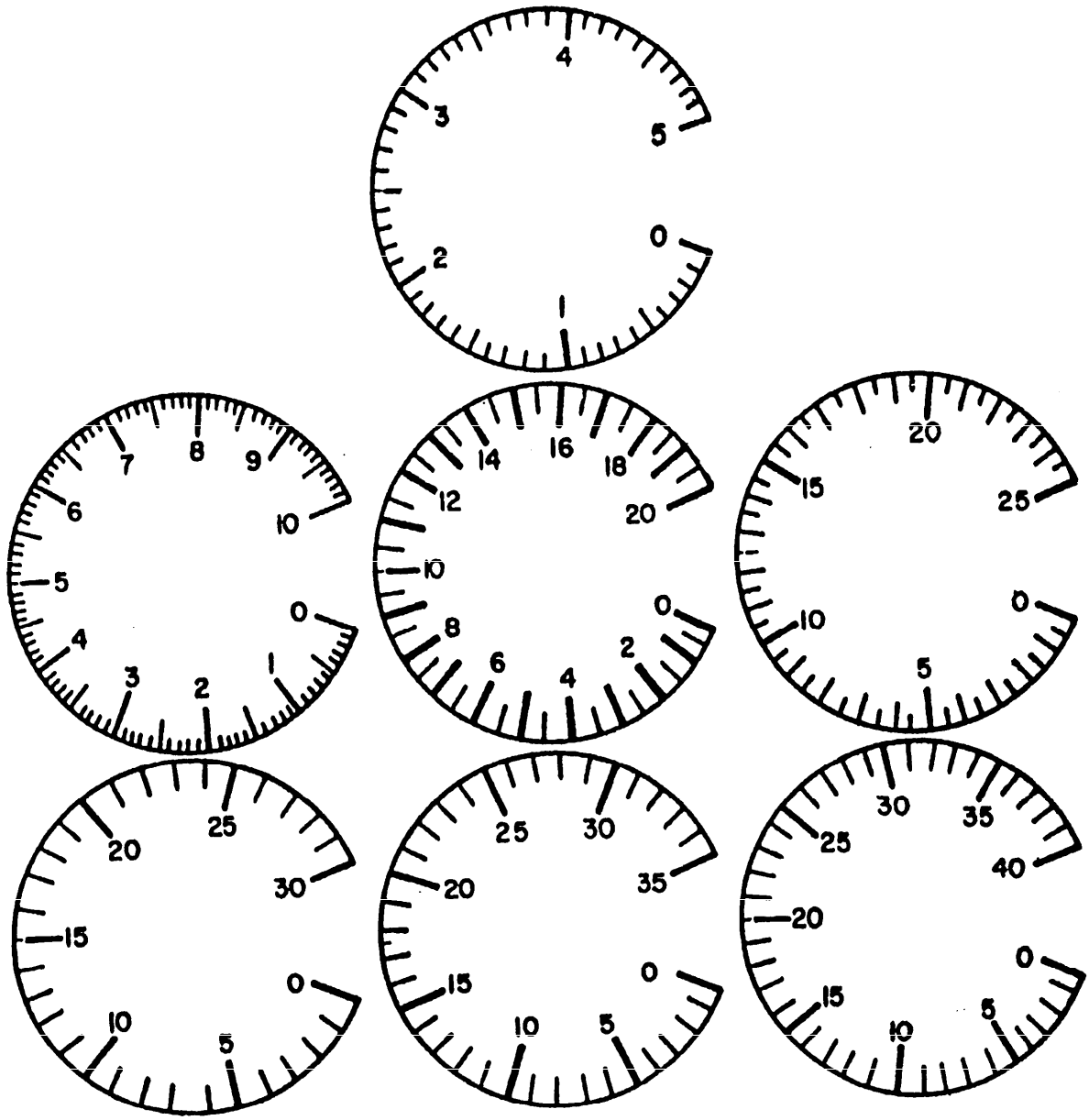


FIGURE 3 SCALES

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joining the appropriate graduation and the center of the dial. Any confusion resulting in doubt as to the graduation to which the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be not less than 1/64th inch. The dimensions, in inches, of the markings on the dial of each indicator shall be as listed in Table I.

TABLE I

DIAL MARKING - INCHES					
	2-Inch Size		3-Inch Size		
MARKING	Length or Height ±0.010	Width of Graduation or Line ±0.005	Length or Height ±0.015	Width of Graduation or Line ±0.010	Finish
Scale arc	----	0.010	----	0.015	
Major graduations	0.188	0.020	0.250	0.030	
Intermediate graduations, if any	0.125	0.015	0.188	0.020	
Minor graduations	0.100	0.010	0.156	0.015	Lusterless
Numerals	0.156	0.015	0.200	0.020	
"PRESSURE"	0.125	0.015	0.150	0.020	White
Electrical zero reference mark	0.031	0.010	0.031	0.015	
Shaded portion of pointers	----	----	----	----	
Background of "OFF" alarm sign	----	----	----	----	
Background of dial	----	----	----	----	Lusterless
Unshaded portion of pointers	----	----	----	----	Black
"OFF"	0.125	0.015	0.150	0.020	

3.6.2.1 Electrical zero reference mark - The electrical zero reference mark shall be located at 1/2 scale pressure indication and shall be in accordance with Table I and Figures 2 and 3. The electrical zero reference mark shall be visible through the cover glass.

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3.6.3 "OFF" alarm - Each indicator shall have an "OFF" alarm sign which shall be visible through the cover glass at all times when no power is applied to the pins of the electric connector of the indicator to which the rotors of the indicator synchros are connected. The "OFF" alarm sign shall be located as shown in Figure 2 and shall be marked in accordance with Table I.

3.6.4 Pointers - The pointers of each indicator shall be marked with the numbers listed for the applicable MS designation (see 1.2). Except for length, the pointer nearest to the dial of the indicator shall conform to MS33585-11 or MS33585-13, as applicable, and the pointer nearest to the cover glass shall conform to MS33585-12 or MS33585-14, as applicable. Each pointer shall overlap each major graduation but shall not overlap any minor graduation.

* 3.7 Cover glass - The cover glass shall be in accordance with Specification DD-G-451, Type II, Quality AA. Any flaws permitted by Specification DD-G-451 shall not interfere with reading of the indicator.

* 3.7.1 Cover glass mounting - The distance from the inner surface of the cover glass to the surface of the dial on which the marking is applied shall not exceed 0.188 inch.

3.8 Performance - The indicator shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

* 3.8.1 Reliability in Mean Time Between Failures - The indicator shall have a minimum of 1000 hours of mean (operating) time between failures when tested for acceptance as outlined under the requirements of 4.4.3.

3.9 Identification of product -

* 3.9.1 Nameplate - A nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130 except that the FSN marking shall be omitted.

3.9.2 Manufacturer's part number - The manufacturer's part number on the name plate shall be identical with the manufacturer's engineering production drawing number, including applicable dash numbers if the drawing is tabulated and covers more than one part.

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3.10 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.11 Workmanship - The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine-screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges.

3.11.1 Dimensions - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. When dimensions and tolerances affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

* 3.11.2 Fabrication - Machining, drilling, and forming shall be done with the use of accurate templates, jigs or gages.

* 3.11.3 Screw assemblies - Assembly screws shall be tight. The word "tight" means that the screws cannot be appreciably tightened without damage or injury to the screw threads or heads.

* 3.11.4 Riveting - Riveting operations shall be performed carefully in order to insure that the rivets are tight. Rivet heads shall be full size and concentric with the rivet body. Dimensions of the rivets shall be such that the holes will be filled. Loose, cracked, or badly formed, including protruding countersunk rivet heads, shall be replaced before acceptance of the indicator. Excess metal shall be removed from countersunk rivet heads as a customary operation. This does not cover eyelets or hollow tubular rivets.

* 3.11.5 Gears - Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Where required for accurate adjustments, gear assemblies shall be as free as possible from backlash.

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* 3.11.6 Cleanup - The indicators shall be thoroughly cleaned of loose, spattered, or excess solder, resin flash which may crumble, metal chips and other foreigh matter, after final assembly.

* 3.11.7 Wire stripping - Wire stripping shall be done without nicking or otherwise damaging the wire.

3.11.8 Soldering - Soldering shall be performed in accordance with Specification MIL-S-6872.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Contractor's responsibility - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

* 4.2 Classification of tests - The inspection and testing of the indicators shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests performed after award of contract on sample indicators to determine that the indicator meets the requirements of this specification. The sample indicators shall be representative in design, performance, and configuration of the indicators which will be produced on the manufacturer's production line.

(b) Acceptance tests: Acceptance tests are those tests performed on indicators manufactured and submitted for acceptance under contract.

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* 4.3 Preproduction tests - The preproduction tests of the indicators shall consist of all of the tests of this specification specified under the paragraph headed Test methods conducted in the order listed.

* 4.3.1 Preproduction test sample instructions - Preproduction test samples shall consist of three indicators manufactured in accordance with this specification. The indicators submitted for preproduction testing shall not have been previously tested except for the Individual tests. The samples shall be forwarded to the testing laboratory designated by the procuring activity.

* 4.3.1.1 Preproduction test sample identification - The test samples shall be plainly identified by securely attached, durable tags marked with the following information:

Sample for Preproduction Testing
Indicators, Pressure, Synchro, Dual, General Purpose
Submitted by (Manufacturer's name, date)
for Preproduction Testing in accordance with
Specification MIL-I-22054A under contract or
order number _____.
MS Part No.
Manufacturer's Part Number

* 4.4 Acceptance tests - The acceptance tests shall consist of the Individual tests, the Sampling plan tests and the Reliability assurance tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests except that the Sampling plan B tests shall be conducted at a Government laboratory designated by the procuring activity. Acceptance tests, except for Sampling plan B tests, shall be under the supervision of the Government inspector. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

* 4.4.1 Individual tests - Each indicator submitted for acceptance shall be subjected to the Individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship,

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operational adequacy, and reliability. As a minimum, each indicator accepted shall have passed the following tests, conducted in the order listed.

Examination of product
Electrical zero
Scale error and friction
Position error
"OFF" alarm
Dielectric strength
Integral lighting (where applicable)

* 4.4.2 Sampling plan tests - The sampling plan tests shall consist of Sampling plan A and Sampling plan B tests. The test samples selected for sampling tests shall first have passed the Individual tests. When an indicator fails to meet the requirements of this specification, indicators still on hand or subsequently produced shall not be accepted until the extent and cause of failure have been determined and corrected. In addition, when a failure occurs, shift to one sample out of the first fifteen (when sampling plan B is omitted) and proceed as indicated in 4.4.2.1. The test samples which have been subjected to Sampling plan A tests shall not be delivered on contract until they have been refurbished, resubmitted and passed all the Individual tests. Test samples which have been subjected to the Sampling plan B tests shall not be delivered on contract.

* 4.4.2.1 Sampling plan A sample selection - Sampling plan A samples shall be selected at random in accordance with the following schedule:

Quantity of Indicators Offered for Acceptance		Quantity to be Selected for Testing
First	15	*
Next	50	1
Next	75	1
Next	100	1
Each additional 200 or fraction thereof		1

*NOTE: When Sampling plan B is invoked the quantity shall be zero. When Sampling plan B is to be omitted the quantity shall be one.

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* 4.4.2.1.1 Sampling plan A tests - Each sample selected for Sampling plan A testing shall be subjected to the following tests, conducted in the order listed:

Magnetic effect
High temperature operation
Low temperature operation
Case leakage
Vibration (resonance)

* 4.4.2.2 Sampling plan B instructions - Two indicators shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These samples shall be forwarded to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached, durable tag marked with the following information:

Indicator, Pressure, Synchro, Dual, General Purpose
Submitted by (Manufacturer's name, date)
for production acceptance Sampling plan B testing, in accordance with Contract or Order No._____.
Manufacturer's part number_____

* 4.4.2.2.1 Sampling plan B tests - Each sample selected for Sampling plan B testing shall be subjected to the following tests, conducted in the order listed:

Sampling plan A
Vibration (cycling)
Radio noise interference
Fogging
High temperature exposure
Low temperature exposure
Effect of lead length
Endurance
Lamp life (where applicable)
Thermal shock
Lighting contrast (where applicable)
Test for helium
Torque
Internal examination

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4.4.3 Reliability tests -

* 4.4.3.1 Reliability assurance tests - Test samples which have successfully passed the Individual tests shall be tested for reliability quality assurance in accordance with Specification MIL-R-23094, Procedure I.

* 4.4.3.2 Test level - The test level shall be Test Level III from Table I of Specification MIL-R-23094. During each cycle, the indicator shall be operated only during the heat portion of the cycle.

* 4.4.3.3 Duty cycle - The duty cycle shall consist of cycling the indicator between approximately 15 and 85 percent of full scale at a rate of 15 ± 5 cycles per minute.

* 4.4.3.4 Accept-reject criteria - Figure 11 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the Reliability Qualification Plan and Figure 12 shall be used for the Reliability Sampling Plan.

* 4.4.3.5 Length of Heat portion of cycle - After stabilization at the high temperature limit required by the test level, the indicator shall be operated 3 hours during each cycle.

* 4.4.3.6 Performance characteristics to be measured -

Characteristic	Specification requirement	Time in cycle that measurement is to be made.
Scale error and friction	Tolerance specified in Table II for high temperature	Immediately prior to cooling portion of cycle but not during the vibration period.
"OFF" alarm	See 4.6.5	Immediately after beginning of heat portion of cycle
Integral lighting	See 4.6.7 (Operation)	Same as for "OFF" alarm

* 4.4.3.7 Failure criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(a) Whenever performance characteristics fall below the acceptance requirements of 4.4.3.6, at least one failure

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has occurred. If subsequent analysis reveals that several parts have deteriorated, each should be counted a failure, unless one part caused the other parts to fail.

* 4.4.3.8 Preventive maintenance - During the period of the tests no preventive maintenance may be performed upon the indicator. Maintenance shall be limited to the correction of failures.

4.5 Test conditions -

* 4.5.1 Standard conditions - Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature	Room ambient 25° ±5° C.
Pressure	Normal atmospheric (approximately 29.92 inches Hg.)
Humidity	Room ambient up to 90 percent relative humidity
Input power	26 ± 0.5V AC, 400 ±20 cps

4.5.2 Test readings - Unless otherwise specified, before a test reading is taken the indicator shall be vibrated using an electric vibrator set at 30 cps, 60 cps, or 120 cps with a maximum amplitude of 0.002 inch.

4.5.3 Attitude - Unless otherwise specified, the indicator shall be tested in its normal operating position.

4.5.4 Electrical zero - The electrical zero (synchro zero) test shall be accomplished by the method specified in Specification MIL-S-20708.

4.5.5 Test standard - The indicator shall be tested with a transmitting standard designed especially for the purpose. Sufficient tests shall be performed to determine that the transmitted signals provide a torque on the indicator of 2200 ±10 percent milligram millimeters per degree, and with a maximum transmitted error not over 0.20 circular degree.

* 4.5.6 Integral lighting - For indicators having integral lighting, the indicator lighting system shall be energized and shall operate satisfactorily during all tests except salt spray and as otherwise specified herein. The lighting shall operate satisfactorily after completion of each test unless otherwise specified herein.

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4.6 Test methods -

4.6.1 Examination of product - Each indicator shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.6.2 Electrical zero - For the purpose of this test, a single indicator with the glass removed shall be connected to the indicator being tested and used as the transmitter unit. The standard transmitter shall not be used. With the power source disconnected, the pointer of the single indicator shall be displaced 180 degrees from the pointer position of the indicator being tested. The pointer of the single indicator shall be held in this position and power shall be applied. This procedure shall be repeated a minimum of three times, after which electrical zero shall be imposed on the indicator being tested. The tip of each pointer shall be within 0.010 inch of its electrical zero reference mark and the tips of the pointers shall be within 0.010 inch of each other.

4.6.3 Scale error and friction - The knob of a standard transmitter shall be rotated to bring its pointer to the desired indication. Two readings shall be taken; the first before, and the second after, the indicator is vibrated. The difference between the first and second readings shall not exceed the friction tolerance specified in Table II at room temperature. The second reading shall not exceed the scale error tolerance specified in Table II at room temperature.

TABLE II

TOLERANCES IN PSI FOR VARIOUS TEST CONDITIONS							
Range	0-5	0-10	0-20	0-25	0-30	0-35	0-40
Scale Error							
Room temperature	0.025	0.050	0.100	0.125	0.150	0.175	0.200
High temperature	0.050	0.100	0.200	0.250	0.300	0.350	0.400
Low temperature	0.050	0.100	0.200	0.250	0.300	0.350	0.400
High temperature exposure	0.050	0.100	0.200	0.250	0.300	0.350	0.400
Low temperature exposure	0.050	0.100	0.200	0.250	0.300	0.350	0.400
Vibration	0.025	0.050	0.100	0.125	0.150	0.175	0.200
Endurance	0.050	0.100	0.200	0.250	0.300	0.350	0.400
Friction							
Room temperature	0.025	0.050	0.100	0.125	0.150	0.175	0.200
High temperature	0.025	0.050	0.100	0.125	0.150	0.175	0.200
Low temperature	0.050	0.100	0.200	0.250	0.300	0.350	0.400
High temperature exposure	0.050	0.100	0.200	0.250	0.300	0.350	0.400

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TABLE II (Continued)

Range	0-5	0-10	0-20	0-25	0-30	0-35	0-40
Friction (continued)							
Low temperature exposure	0.050	0.100	0.200	0.250	0.300	0.350	0.400
Position error at 1/2 scale	0.025	0.050	0.100	0.125	0.150	0.175	0.200
Vibration pointer oscillation	0.025	0.050	0.100	0.125	0.150	0.175	0.200
Endurance	0.050	0.100	0.200	0.250	0.300	0.350	0.400

* 4.6.4 Position error - The indicator shall be held in the normal operating position and a reading taken. It shall then be vibrated in accordance with 4.5.2 and a second reading taken. The indicator shall then be held in the following positions:

(a) Rotated from the normal operating position through 45 degrees to the rear (dial tilted 45 degrees backward from the vertical position).

(b) Rotated from the normal operating position through 45 degrees to the front (dial titled 45 degrees forward from the vertical position).

(c) Rotated from the normal operating position through 45 degrees to the right (dial vertical).

(d) Rotated from the normal operating position through 45 degrees to the left (dial vertical).

The maximum deviation of the four latter readings, from the original reading, shall not exceed the tolerance specified in Table II.

4.6.5 "OFF" alarm - A normal excitation voltage of 26 volts, 400 cycles shall be applied to the pins of the electric connector of the indicator to which the rotors of the indicator synchros are connected. The "OFF" alarm sign shall not appear while the voltage is applied. The voltage shall then be removed and the "OFF" alarm sign shall appear while no voltage is applied.

4.6.6 Dielectric strength - A potential of 200 volts (root mean square) alternating current at commercial frequency shall be applied between isolated pins and the case for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the indicator as a result of this test.

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* 4.6.7 Integral lighting - Each indicator having integral lighting shall be subjected to and shall meet the requirements of the following tests specified in Specification MIL-L-25467 except as indicated herein:

1. Operation
2. Dial visibility
3. Dielectric strength (200 V)

4.6.8 Magnetic effect - The indicator shall be properly connected and power applied. The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees. At each of these positions the indicator shall be rotated 360 degrees about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed one degree. This test shall be repeated with no power applied.

4.6.9 High temperature operation - (This test may be combined with the High temperature exposure test at the discretion of the testing agency.) The indicator shall be properly connected except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of 71° C (160° F) for a period of 4 hours. During the last 5 minutes of the 4-hour period, power shall be applied to the indicator. At the end of the 4-hour period and with the temperature maintained at 71° C(160° F), the indicator shall be subjected to and shall meet the requirements specified in the following tests:

1. Scale error and friction. The tolerances shall be as specified in Table II for high temperature.
2. "OFF" alarm.
3. Upon return to standard conditions, the indicator shall be subjected to and shall meet the requirements of 4.6.3 and 4.6.5.

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4.6.10 Low temperature operation - (This test may be combined with the Low temperature exposure test at the discretion of the testing agency.) The indicator shall be properly connected except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of -54° C (-65° F) for a period of 4 hours. During the last 5 minutes of the 4-hour period power shall be applied to the indicator. At the end of the 4-hour period and with the temperature maintained at -54° C (-65° F), the indicator shall be subjected to and shall meet the requirements specified in the following tests:

1. Scale error and friction. The tolerances shall be as specified in Table II for low temperature and each indicator shall not require more than 5 seconds to come within the tolerances.

2. "OFF" alarm.

3. Upon return to standard conditions, the indicator shall be subjected to and shall meet the requirements of 4.6.3 and 4.6.5.

4.6.11 Case leakage - Each indicator shall be tested for case leakage using a mass spectrometer or equally sensitive quantitative leak test procedure. The gas or mixture of gasses in the case, adjusted to a pressure differential of one atmosphere, shall not have a leakage rate from the case exceeding one micron cubic foot per hour (10^{-5} cubic centimeters per second).

* 4.6.12 Vibration (resonance) - The indicator shall be tested in accordance with Procedure XIII of Specification MIL-E-5272. During the vibration, the "OFF" alarm and integral lighting (where applicable) shall operate properly and the scale error and pointer oscillation shall not exceed the tolerances specified in Table II.

* 4.6.13 Vibration (cycling) - The indicator shall be tested in accordance with Procedure XIII of Specification MIL-E-5272. Upon completion of the vibration, the indicator shall be subjected to and shall meet the requirements of the following tests:

1. Scale error and friction. The tolerances shall be as specified in Table II for room temperature.

2. "OFF" alarm.

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No parts shall become loose nor shall damage to any part of the indicator result from this test.

4.6.14 Radio noise interference - The electrical system of the indicator shall be tested for conducted and radiated radio noise interference in accordance with Specification MIL-I-6181.

* 4.6.15 Fogging - The indicator shall be energized and placed in a 71° ±2°C controlled ambient temperature for a minimum period of one hour. After this period, and while still at this temperature, an ice cube shall be rubbed on the cover glass for a period of from one to two minutes. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog inside the cover glass. Evidence of fogging shall be cause for rejection.

4.6.16 High temperature exposure - The indicator shall be tested in accordance with Procedure II of Specification MIL-E-5272. After completion of test, the indicator shall be subjected to and shall meet the requirements specified in the following tests:

1. Scale error and friction. The tolerances shall be as specified in Table II for high temperature and each indicator shall not require more than 5 seconds to come within the tolerances.
2. "OFF" alarm.
3. Upon return to standard conditions, the indicator shall be subjected to and shall meet the requirements of 4.6.3 and 4.6.5.

4.6.17 Low temperature exposure - The indicator shall be tested in accordance with Procedure II of Specification MIL-E-5272. After completion of the test the indicator shall be subjected to and shall meet the requirements specified in the following tests:

1. With the temperature at -65° C (-80° F), the indicator shall be checked to assure that the pointer moves smoothly over the entire scale range, and to determine that no component parts have been damaged.
2. With the temperature at -54° C (-65° F), the indicator shall be subjected to the Scale error and friction test except that in every case the indicator shall not require

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more than 30 seconds to come within the tolerances specified in Table II for low temperature exposure.

3. Upon return to standard conditions, the indicator shall be subjected to and shall meet the requirements of 4.6.3 and 4.6.5.

* 4.6.18 Effect of lead length - The indication shall not change more than 1 degree of an arc when the lead length between indicator and transmitter is changed from 5 to 500 feet of No. 20 AWG gage wire. During this test, the power ground wire, Pin A, shall be grounded at both the indicator and transmitter to simulate aircraft ground.

* 4.6.19 Endurance - The indicator pointer shall be caused to rotate at a rate of 60 ±10 cycles per minute for a period of 100,000 cycles. The first 50,000 cycles shall be in the clockwise direction, the second 50,000 cycles shall be in the counter-clockwise direction. At the end of this period the indicator shall be subjected to and shall meet the requirements specified in the following tests:

1. Scale error and friction. The tolerance shall be as specified in Table II for endurance.

2. "OFF" alarm.

* 4.6.20 Lamp life - The lighted indicator shall be subjected to the life test specified in Specification MIL-L-25467. (This test shall be run concurrently with the Endurance test, if practicable.)

4.6.21 Salt spray - The indicator shall be tested in accordance with Procedure I of Specification MIL-E-5272. After completion of test the indicator shall be subjected to and shall meet the requirements specified in the following tests:

1. Dielectric strength

2. Case leakage

* 4.6.22 Thermal shock - The indicator shall be subjected to eight cycles of exposure to water at 85° ±2° C (185°±3.6° F) and 5° ±2° C (41° ±3.6° F) without evidence of moisture penetration or damage to coating or enclosure. Each cycle of the test shall consist of immersing the indicator in water at 85° C

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for a period of 25 minutes, and then, within 5 seconds of removal from the bath, the indicator shall be immersed for a period of 25 minutes in the other water bath maintained at 5° C. This cycle shall be repeated continuously, one cycle following the other, until eight cycles have been completed. Upon completion of this test, the indicator shall be subjected to and shall meet the requirements of the case leakage test.

* 4.6.23 Lighting contrast - The lighted indicator shall be subjected to and shall meet the requirements of the contrast test specified in Specification MIL-L-25467.

* 4.6.24 Test for helium - The indicator case shall be punctured, or the filling tube cut, and the indicator shall be subjected to the test for helium by means of a mass spectrometer type helium detector. Failure to detect helium shall be cause for rejection.

4.6.25 Torque - The test shall be accomplished with the three secondary leads of one indicator synchro connected to the corresponding leads of a transmitter synchro conforming to Specification MIL-T-5350. A normal excitation of 26 volts, 400 cps shall be supplied to the primaries (rotors) of both synchros. The rotor of the transmitter synchro shall be locked in one position and the rotor of the indicator synchro rotated through an angle from the corresponding synchronous position. The torque tending to return the rotor of the indicator synchro to its synchronous position shall be measured. The torque is proportional to the angular displacement within displacements up to 20 degrees and shall be no less than 2,000 milligram millimeters per degree of displacement.

* 4.6.26 Internal examination - The case of the indicator shall be opened and the internal mechanism shall be examined. The lamp circuit of the indicator shall also be examined as specified in Specification MIL-L-25467. Any deterioration or damage which could in any manner prevent the indicator from meeting functional operation or maintenance requirements during service life shall be cause for rejection.

4.7 Rejection and retest - Indicators which have been rejected may be reworked or have parts replaced to correct the defects and may be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and action taken to correct the defects shall be furnished the Government inspector.

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5. PREPARATION FOR DELIVERY

5.1 Packaging - The indicators shall be packaged in accordance with Specification MIL-P-7936, Level A or C. When Level B is required, the details of its accomplishment shall be specified in the contract or order (see 6.2). For Level A packaging the method of preservation shall be in accordance with MIL-P-116, Method 1A-5 without preservative compound, using metal reusable containers.

5.2 Packing - The indicators shall be packed in accordance with Specification MIL-P-7936, Level A, B or C as specified in contract or order (see 6.2).

5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

5.3.1 Precautionary Marking - The following precautionary markings shall appear on two opposite sides of each interior package and shipping container whenever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use - The indicators covered by this specification are intended for use on aircraft to provide remote indication of aircraft pressures.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, Number and Date of this Specification.

(b) The quantity and MS part number of indicators desired.

(c) Levels of packaging (see 5.1) and packing (see 5.2) desired.

6.3 Drawing requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327.

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6.4 Definitions -

6.4.1 Amplitude - Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.4.2 Hermetic seal - A hermetic seal shall be a perfectly closed and air tight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals accomplished by gaskets.

* 6.5 Preproduction test provisions - The manufacture of indicators on contract shall not commence until the samples submitted are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of the indicator from contract to contract, the submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirement of acceptance testing.

6.5.1 It shall be understood that the indicator supplied under contract or order shall be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship and method of manufacture. Deviation from the standards of the preproduction sample shall be made only by the procuring activity. Evidence of unauthorized change shall constitute cause for rejection.

* 6.6 The outside margins of this specification have been marked where changes from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

6.7 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents referenced in Section 2, unless specified in the contract or order.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-I-22054A(WP) INDICATORS, PRESSURE, SYNCHRO, DUAL, GENERAL PURPOSE

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Bureau of Naval Weapons
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